THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Lines and Angles** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 240	The two angles between the hands add up to 360°.
2	**(d)** 120	6 degrees per minute, or 30° per number on the dial.
3	**(c)** 5	30° per hour.
4	**(a)** 160	Full turn is 360°.
5	**(c)** 27	A bisector divides the angle into two equal parts.
6	**(d)** 270	Add the two turns in degrees.
7	**(c)** right	Find the angle first.
8	**(a)** 75	Straight angle is 180°.
9	**(b)** 210	The two angles between the hands add up to 360°.
10	**(d)** 210	6 degrees per minute, or 30° per number on the dial.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Smaller angle $= 150^\circ$ (obtuse) (1); Larger angle $= 360 - 150 = 210^\circ$ (reflex) (1).	1+1
12	Bisector gives two equal angles (1); $\angle AOC = 2 \times 28 = 56^\circ$, type acute (1).	1+1
13	Hour hand: $4 \times 30 = 120^\circ$ (1); Minute hand: 4 full turns $= 1440^\circ$ (1).	1+1
14	Smaller angle $= 120^\circ$ (obtuse) (1); Larger angle $= 360 - 120 = 240^\circ$ (reflex) (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Three-quarter turn $= 270^\circ$ and quarter turn $= 90^\circ$ (1); Net turn $= 270 - 90 = 180^\circ$ (1); Type: straight (1).	1+1+1
16	Anticlockwise total $= 60 + 70 = 130^\circ$ (1); Clockwise 40° is subtracted (1); Net turn $= 90^\circ$ anticlockwise (right) (1).	1+1+1
17	Total segments $= \frac{5\times4}{2} = 10$ (1); Segments having A as an end point: 4 (1); Segments without A: $10 - 4 = 6$ (1).	1+1+1
	OR: Full turn $= 360^\circ$ and quarter turn $= 90^\circ$ (1); Net turn $= 360 - 90 = 270^\circ$ (1); Type: reflex (1).	1+1+1
18	Anticlockwise total $= 45 + 100 = 145^\circ$ (1); Clockwise 85° is subtracted (1); Net turn $= 60^\circ$ anticlockwise (acute) (1).	1+1+1
	OR: Total segments $= \frac{6\times5}{2} = 15$ (1); Segments having A as an end point: 5 (1); Segments without A: $15 - 5 = 10$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Hours passed $= 2$, hour hand turns 60° (1); Minute hand turns 2 full turns $= 720^\circ$ (1); At 3:00 the hour hand points to 3 ($= 90^\circ$) (1); Smaller angle between the hands $= 90^\circ$ (1); Type: right (1).	1+1+1+1+1
	OR: $\angle AOD = 30 + 45 + 60 = 135^\circ$ (1); Type: obtuse (1); Angle left $= 180 - 135 = 45^\circ$ (1); Check with the sum of the parts (2).	1+1+1+2
20	Hours passed $= 5$, hour hand turns 150° (1); Minute hand turns 5 full turns $= 1800^\circ$ (1); At 7:00 the hour hand points to 7 ($= 210^\circ$) (1); Smaller angle between the hands $= 150^\circ$ (1); Type: obtuse (1).	1+1+1+1+1
	OR: $\angle AOD = 40 + 50 + 70 = 160^\circ$ (1); Type: obtuse (1); Angle left $= 180 - 160 = 20^\circ$ (1); Check with the sum of the parts (2).	1+1+1+2

Mistakes to avoid:

- Reading the wrong scale on the protractor (inner vs outer). Decide first whether the angle is acute or obtuse.
- Using 30° per minute for the minute hand; it is 6° per minute, and 30° per hour is for the hour hand.
- Calling a 200° or 225° angle obtuse. Obtuse angles are less than 180°; larger ones are reflex.
- Saying a ray has two end points; a ray has exactly one end point.
- Counting line segments for n points as $n(n-1)$ without dividing by 2 (segment AB and BA are the same).
- Placing the protractor's centre off the vertex, or starting the scale from the wrong arm.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 2: **Lines and Angles** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 7	30° per hour.
2	**(b)** 120	Full turn is 360°.
3	**(a)** 38	A bisector divides the angle into two equal parts.
4	**(a)** 210	Add the two turns in degrees.
5	**(c)** straight	Find the angle first.
6	**(c)** 90	Straight angle is 180°.
7	**(a)** 210	The two angles between the hands add up to 360°.
8	**(a)** 240	6 degrees per minute, or 30° per number on the dial.
9	**(c)** 9	30° per hour.
10	**(b)** 75	Full turn is 360°.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Bisector gives two equal angles (1); $\angle AOC = 2 \times 47 = 94^\circ$, type obtuse (1).	1+1
12	Hour hand: $5 \times 30 = 150^\circ$ (1); Minute hand: 5 full turns $= 1800^\circ$ (1).	1+1
13	Smaller angle $= 60^\circ$ (acute) (1); Larger angle $= 360 - 60 = 300^\circ$ (reflex) (1).	1+1
14	Bisector gives two equal angles (1); $\angle AOC = 2 \times 65 = 130^\circ$, type obtuse (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Full turn $= 360^\circ$ and one-third turn $= 120^\circ$ (1); Net turn $= 360 - 120 = 240^\circ$ (1); Type: reflex (1).	1+1+1
16	Anticlockwise total $= 90 + 120 = 210^\circ$ (1); Clockwise 30° is subtracted (1); Net turn $= 180^\circ$ anticlockwise (straight) (1).	1+1+1
17	Total segments $= \frac{7\times6}{2} = 21$ (1); Segments having A as an end point: 6 (1); Segments without A: $21 - 6 = 15$ (1).	1+1+1
	OR: Three-quarter turn $= 270^\circ$ and one-sixth turn $= 60^\circ$ (1); Net turn $= 270 - 60 = 210^\circ$ (1); Type: reflex (1).	1+1+1
18	Anticlockwise total $= 35 + 75 = 110^\circ$ (1); Clockwise 50° is subtracted (1); Net turn $= 60^\circ$ anticlockwise (acute) (1).	1+1+1
	OR: Total segments $= \frac{8\times7}{2} = 28$ (1); Segments having A as an end point: 7 (1); Segments without A: $28 - 7 = 21$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Hours passed $= 4$, hour hand turns 120° (1); Minute hand turns 4 full turns $= 1440^\circ$ (1); At 8:00 the hour hand points to 8 ($= 240^\circ$) (1); Smaller angle between the hands $= 120^\circ$ (1); Type: obtuse (1).	1+1+1+1+1
	OR: $\angle AOD = 25 + 55 + 90 = 170^\circ$ (1); Type: obtuse (1); Angle left $= 180 - 170 = 10^\circ$ (1); Check with the sum of the parts (2).	1+1+1+2
20	Hours passed $= 7$, hour hand turns 210° (1); Minute hand turns 7 full turns $= 2520^\circ$ (1); At 10:00 the hour hand points to 10 ($= 300^\circ$) (1); Smaller angle between the hands $= 60^\circ$ (1); Type: acute (1).	1+1+1+1+1
	OR: $\angle AOD = 35 + 60 + 45 = 140^\circ$ (1); Type: obtuse (1); Angle left $= 180 - 140 = 40^\circ$ (1); Check with the sum of the parts (2).	1+1+1+2

Mistakes to avoid:

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- Calling a 200° or 225° angle obtuse. Obtuse angles are less than 180°; larger ones are reflex.
- Saying a ray has two end points; a ray has exactly one end point.
- Counting line segments for n points as $n(n-1)$ without dividing by 2 (segment AB and BA are the same).
- Placing the protractor's centre off the vertex, or starting the scale from the wrong arm.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Lines and Angles** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 44	A bisector divides the angle into two equal parts.
2	**(d)** 360	Add the two turns in degrees.
3	**(b)** acute	Find the angle first.
4	**(d)** 60	Straight angle is 180°.
5	**(a)** 240	The two angles between the hands add up to 360°.
6	**(d)** 300	6 degrees per minute, or 30° per number on the dial.
7	**(d)** 11	30° per hour.
8	**(a)** 50	Full turn is 360°.
9	**(b)** 56	A bisector divides the angle into two equal parts.
10	**(c)** 450	Add the two turns in degrees.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Hour hand: $6 \times 30 = 180^\circ$ (1); Minute hand: 6 full turns $= 2160^\circ$ (1).	1+1
12	Smaller angle $= 30^\circ$ (acute) (1); Larger angle $= 360 - 30 = 330^\circ$ (reflex) (1).	1+1
13	Bisector gives two equal angles (1); $\angle AOC = 2 \times 95 = 190^\circ$, type reflex (1).	1+1
14	Hour hand: $3 \times 30 = 90^\circ$ (1); Minute hand: 3 full turns $= 1080^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Full turn $= 360^\circ$ and one-eighth turn $= 45^\circ$ (1); Net turn $= 360 - 45 = 315^\circ$ (1); Type: reflex (1).	1+1+1
16	Anticlockwise total $= 110 + 80 = 190^\circ$ (1); Clockwise 65° is subtracted (1); Net turn $= 125^\circ$ anticlockwise (obtuse) (1).	1+1+1
17	Total segments $= \frac{9\times8}{2} = 36$ (1); Segments having A as an end point: 8 (1); Segments without A: $36 - 8 = 28$ (1).	1+1+1
	OR: Three-quarter turn $= 270^\circ$ and half turn $= 180^\circ$ (1); Net turn $= 270 - 180 = 90^\circ$ (1); Type: right (1).	1+1+1
18	Anticlockwise total $= 25 + 125 = 150^\circ$ (1); Clockwise 70° is subtracted (1); Net turn $= 80^\circ$ anticlockwise (acute) (1).	1+1+1
	OR: Total segments $= \frac{10\times9}{2} = 45$ (1); Segments having A as an end point: 9 (1); Segments without A: $45 - 9 = 36$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Hours passed $= 6$, hour hand turns 180° (1); Minute hand turns 6 full turns $= 2160^\circ$ (1); At 11:00 the hour hand points to 11 ($= 330^\circ$) (1); Smaller angle between the hands $= 30^\circ$ (1); Type: acute (1).	1+1+1+1+1
	OR: $\angle AOD = 20 + 70 + 80 = 170^\circ$ (1); Type: obtuse (1); Angle left $= 180 - 170 = 10^\circ$ (1); Check with the sum of the parts (2).	1+1+1+2
20	Hours passed $= 3$, hour hand turns 90° (1); Minute hand turns 3 full turns $= 1080^\circ$ (1); At 9:00 the hour hand points to 9 ($= 270^\circ$) (1); Smaller angle between the hands $= 90^\circ$ (1); Type: right (1).	1+1+1+1+1
	OR: $\angle AOD = 50 + 40 + 85 = 175^\circ$ (1); Type: obtuse (1); Angle left $= 180 - 175 = 5^\circ$ (1); Check with the sum of the parts (2).	1+1+1+2

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